

Final

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Problem 1. *What should we mean by The Group of symmetries of X ? i.e. define what we should mean by an automorphism of X , as a G -set.*

Proof. Let X be a G -set and let $n = |X|$. An automorphism of a **set** is any rearrangement, or permutation, of the set. An automorphism of a **G -set** is any rearrangement, or permutation, of the set which ensures the G -set structure is maintained. So $\forall \phi \in \text{Aut}_G(X)$, ϕ is a automorphism, or rearrangement, of X , but it also must ensure the G -set structure is maintained and thus $\forall g \in G$, $\phi(g*x) = g*\phi(x)$. The action by g induces a permutation to X through the homomorphism $f : G \rightarrow S_n$ caused by the action of G defined on X . So $\forall g \in G$, g induces a permutation on X which commutes with the permutation induced by ϕ by the G -set homomorphism rule. Thus $\text{Aut}_G(X)$ is the set of permutations in S_n which commutes with all permutations of action by G induced by f . In other words, $\text{Aut}_G(X)$ is the set of permutations of X which commute with all permutations in the image of G through f . $\square$

Problem 2a. *Explain (briefly) how G acts on $\text{Pow}(G)$ (the set of subsets) by both left, and right multiplication.*

Proof. Let $*$ denote left multiplication by G and let $\cdot$ denote right multiplication by G .

G acts on $\text{Pow}(G)$ by left multiplication by the left multiplication of every element in the subset of G . Similarly G acts on $\text{Pow}(G)$ by right multiplication by the right multiplication of every element in the subset of G . $\forall g \in G$, and $\forall H \in \text{Pow}(G)$, $H = \{h_1, \dots, h_n\}$ where $n = |H|$ and $h_1, \dots, h_n \in G$. $g * H = g * \{h_1, \dots, h_n\} = \{gh_1, \dots, gh_n\}$. $g \cdot H = g \cdot \{h_1, \dots, h_n\} = \{h_1g^{-1}, \dots, h_ng^{-1}\}$. $\square$

Problem 2b. *Now let G/H be the set of cosets $G/H := \{gH | g \in G\}$, a subset of $\text{Pow}(G)$. Show that the right action of $N(H)$ on $\text{Pow}(G)$ preserves G/H - in particular $N(H)$ acts by right multiplication on G/H .*

Proof. $\forall n \in N(H)$ and $\forall gH \in G/H$, $n \cdot gH = gHn^{-1}$. Because $n \in N(H)$ and the normalizer of a subgroup is also a subgroup $n^{-1} \in N(H)$. By definition of a normalizer, $n^{-1}Hn = H \implies n^{-1}H = Hn^{-1}$. Thus $gHn^{-1} = gn^{-1}H \in G/H$. Furthermore, let $n \in N(H)$ and $\forall g \in G$, $g = ge = gnn^{-1}$, thus $gH = gnn^{-1}H = n \cdot gnH$. Thus right action of $N(H)$ on $\text{Pow}(G)$ preserves G/H . $\square$

Problem 2c. *Show that the left action of G on $\text{Pow}(G)$ preserves G/H , in particular G acts by left multiplication on G/H .*

Proof. $\forall g \in G$ and $\forall xH \in G/H$. $g * xH = gxH \in G/H$. $\forall x \in G$, $x = gg^{-1}x$ thus $xH = gg^{-1}xH = g * g^{-1}xH$ $\square$

Problem 2d. Show that the right action of $N(H)$ and left action of G , both on G/H , commute.

Proof. $\forall n \in N(H), \forall g \in G, \forall xH \in G/H, g * (n \cdot xH) = g * xn^{-1}H = gxn^{-1}H = n \cdot gxH = n \cdot g * xH$. Thus the right action of $N(H)$ commutes with the left action of G on G/H $\square$

Problem 3. Determine (with proof) $\text{Aut}_G(G/H)$ – automorphisms of this G -set as in 1. A correct answer will relate (in a precise way) $\text{Aut}_G(G/H)$ to $N(H)$.

Proof. Let $\alpha : N(H) \rightarrow \text{Aut}_G(G/H)$ defined as the following $\forall n \in N(H), \alpha(n) = \phi_n$ where $\forall xH \in G/H, \phi_n(xH) = n \cdot xH = xn^{-1}H$.

1. **Homomorphism:** $n_1, n_2 \in N(H), \alpha(n_1 n_2) = [xH \rightarrow xn_2^{-1}n_1^{-1}H] = [xH \rightarrow xn_1^{-1}H] \circ [xH \rightarrow xn_2^{-1}H] = \phi_{n_1} \circ \phi_{n_2} = \alpha(n_1) \circ \alpha(n_2)$
2. **Surjective:** $\forall f \in \text{Aut}_G(G/H), f(H) = xH, x \in G. \forall h \in H, xH = f(H) = f(hH)$. Since f is a automorphism of G -sets, $f(hH) = hf(H)$. Thus $xH = hxH$. $x^{-1}xH = H = x^{-1}hxH$. Thus $\forall h \in H, x^{-1}hx \in H$. Thus $x^{-1}Hx \subseteq H$. But since $x^{-1}Hx \cong H$ then $|x^{-1}Hx| = |H|$ and $x^{-1}Hx = H$. Thus by definition of normalizer, $x \in N(H)$. Thus $f = \phi_{x^{-1}} = \alpha(x^{-1})$. Thus α is surjective.

$\forall k \in \ker \alpha, \alpha(k) = \phi_k = id, \phi_k(H) = k^{-1}H = H$. Thus $k^{-1} \in H$ thus $k \in H$. Thus $\ker \alpha \subseteq H$. $\forall h \in H, \alpha(h) = \phi_h, \phi_h(H) = h^{-1}H = H$, thus $\forall xH \in G/H, \phi_h(xH) = xh^{-1}H = xH$. Thus $\phi_h = id$. Thus $h \in \ker \alpha \implies H \subseteq \ker \alpha \implies H = \ker \alpha$. By First Isomorphism Theorem since α is a surjective homomorphism then $\text{Aut}_G(G/H) \cong N(H)/\ker \alpha = N(H)/H$. Thus $\text{Aut}_G(G/H) \cong N(H)/H$ $\square$

Problem 4. Let X and Y be transitive G -sets (ie G acts on either with a single orbit). Show they are isomorphic, as G -sets iff we can find x in X, y in Y , with equal stabilizers, ie $G^x = G^y$, and that this holds iff G^x and G^y are conjugate subgroups of G for any x in X and any y in Y .

Claim 4a. Show $\exists x \in X, y \in Y$ with equal stabilizers, ie $G^x = G^y$, and that this holds iff G^x and G^y are conjugate subgroups of G for any x in X and any y in Y .

Proof. Let X and Y be transitive G -sets.

$\implies$: Suppose $\exists x \in X, y \in Y \ni G^x = G^y$. Since X and Y are transitive, $\forall a \in X$ and $\forall b \in Y$, $\exists g, h \in G, a = gx$ and $b = hy$. By the conjugation formula for stabilizers in the same orbit, we know $G^a = G^{gx} = gG^xg^{-1}$ and $G^b = G^{hy} = hG^yh^{-1}$. Thus $G^a = g^{-1}G^xg = G^y$. Thus $G^b = hg^{-1}G^a gh^{-1} = (hg^{-1})G^a(hg^{-1})^{-1}$ thus G^a and G^b are conjugate subgroups. Thus $\forall a \in X$ and $\forall b \in Y, G^a$ is a conjugate subgroup to G^b .

$\impliedby$: Suppose G^x is conjugate to G^y for any $x \in X$ and any $y \in Y$. Thus by definition, $\exists g \in G$ s.t $G^y = gG^xg^{-1}$ for some $a \in X$ and $y \in Y$. $ga = x \in X$ by definition of group action. By the conjugation formula for stabilizers in the same orbit, we know $G^x = G^{ga} = gG^a g^{-1} = G^y$. Thus $\exists x \in X, y \in Y \ni G^x = G^y$. $\square$

Claim 4b. Show they are isomorphic, as G -sets iff we can find x in X, y in Y , with equal stabilizers, ie $G^x = G^y$

Proof. $\implies$: $X \cong Y$ as G -sets. Since X and Y are transitive and only have a single orbit, $X = G * x$ and $Y = G * y \forall x \in X$ and $\forall y \in Y$. Since $G * x \cong G/G^x$ and $G * y \cong G/G^y$, thus $G/G^x \cong G/G^y$. Thus $G^x \cong G^y$ thus G^x and G^y are conjugate subgroups. Thus by the previous claim, $\exists x \in X$ and $\exists y \in Y$ such that $G^x = G^y$.

$\Leftarrow$: Suppose $\exists x \in X, y \in Y \ni G^x = G^y$. Since X and Y are transitive, $X = G * x$ and $Y = G * y$. Since $G * x \cong G/G^x$ and $G * y \cong G/G^y$ as G -sets. Since $G^x = G^y$, then $X = G * x \cong G/G^x = G/G^y \cong G * y = Y$. Thus $G * x \cong G * y$ and $X \cong Y$. $\square$

Problem 5. Let X be a G -set. Construct a natural homomorphism $\text{Aut}_G(X) \rightarrow \text{Perm}(X/G)$. Describe, with proof, the kernel.

Proof. Let $f : \text{Aut}_G(X) \rightarrow \text{Perm}(X/G)$, defined as the follows, $\forall \phi \in \text{Aut}_G(X)$, $f(\phi) = [G * x \rightarrow \phi_*(G * x)]$.

$\forall \phi_1, \phi_2 \in \text{Aut}_G(X)$, $f(\phi_1 \circ \phi_2) = [G * x \rightarrow \phi_{1*}(\phi_{2*}(G * x))] = [G * x \rightarrow \phi_{1*}(G * x)] \circ [G * x \rightarrow \phi_{2*}(G * x)] = f(\phi_1) \circ f(\phi_2)$. Thus f is a homomorphism.

Let $X/G = \{G * x_1, \dots, G * x_n\}$ be the set of all the distinct orbits of X .

Let $K = \ker f$.

$\forall i \in \{1, \dots, n\}$, let $g : \text{Aut}_G(G * x_i) \rightarrow \text{Aut}_G(X)$ such that $\phi \rightarrow \bar{\phi}$ where $\forall x \in X$, if $x \in G * x_i$ $\bar{\phi}(x) = \phi(x)$ else $\bar{\phi}(x) = id_X(x) = x$.

$\forall a, b \in \text{Aut}_G(G * x_i)$, $\forall x \in X$, if $x \in G * x_i$ $g(a \circ b)(x) = (a \circ b)(x) = (\bar{a} \circ \bar{b})(x) = (g(a) \circ g(b))(x)$ and if $x \notin G * x_i$ then $g(a \circ b)(x) = id_X(x) = (id_X \circ id_X)(x) = (g(a) \circ g(b))(x)$. Thus g is a homomorphism. $\forall k \in \ker g$, $\forall x \in X$, if $x \notin G * x_i$ $g(k)(x) = x$ by definition. If $x \in G * x_i$, $g(k)(x) = k(x) = x$. Thus $\forall x \in G * x_i$, $k(x) = x$ and the only automorphism which fixes all elements of a G -set is the identity. thus $k = id_{G * x_i}$. Thus g is injective and $\text{Aut}_G(G * x_i) \leq \text{Aut}_G(X)$ naturally.

$\forall i \in \{1, \dots, n\}$, $\forall \phi_i \in \text{Aut}_G(G * x_i)$. $\forall x \in G * x_i$, by definition $\bar{\phi}_j(x) = id_X(x) = x$ if $i \neq j$. By definition, $\bar{\phi}_i(x) \in G * x_i$. $(\bar{\phi}_1 \circ \dots \circ \bar{\phi}_n)(x) = \phi_i(x) \in G * x_i$. Thus since each of the orbits is preserved, $(\bar{\phi}_1 \circ \dots \circ \bar{\phi}_n) \in K$.

Let $h : \text{Aut}_G(G * x_1) \times \dots \times \text{Aut}_G(G * x_n) \rightarrow K$ where $(\phi_1, \dots, \phi_n) \rightarrow \bar{\phi}_1 \circ \dots \circ \bar{\phi}_n$. $\forall i, j \in \{1, \dots, n\} \ni i \neq j$, $\forall a \in \text{Aut}_G(G * x_i)$ and $\forall b \in \text{Aut}_G(G * x_j)$. $\forall x \in X$:

1. If $x \notin G * x_i$ and $x \notin G * x_j$, then $(\bar{a} \circ \bar{b})(x) = \bar{a}(id_X(x)) = id_X(x) = x$ and $(\bar{b} \circ \bar{a})(x) = \bar{b}(id_X(x)) = x$. Thus if $x \notin G * x_i \cup G * x_j$, $(\bar{a} \circ \bar{b})(x) = (\bar{b} \circ \bar{a})(x)$
2. If $x \in G * x_i$. x cannot be in $G * x_j$ because the orbits are pairwise disjoint and if $x \in G * x_j$ then $G * x_i = G * x_j \implies i = j$ which is a contradiction with our assumption. $(\bar{a} \circ \bar{b})(x) = a(x)$. $(\bar{b} \circ \bar{a})(x) = \bar{b}(a(x)) = id_X(a(x)) = a(x)$. Thus if $x \in G * x_i$, $(a \circ b)(x) = a(x) = (b \circ a)(x)$
3. If $x \in G * x_j$. Then $(\bar{a} \circ \bar{b})(x) = \bar{a}(b(x)) = b(x)$ and $(\bar{b} \circ \bar{a})(x) = b(x)$. Thus if $x \in G * x_j$, $(a \circ b)(x) = b(x) = (b \circ a)(x)$

Thus $\forall x \in X$, $(a \circ b)(x) = (b \circ a)(x) \implies ab = ba$. Thus $\text{Aut}_G(G * x_i) \text{Aut}_G(G * x_j) = \text{Aut}_G(G * x_j) \text{Aut}_G(G * x_i) \forall i, j \in \{1, \dots, n\} \ni i \neq j$. Thus each pair of subgroups in the direct product commutes in K . Thus all subgroups in the direct product commute. Thus h is a homomorphism.

$\forall k \in K$, $\forall i \in \{1, \dots, n\}$, k preserves each orbit and is a automorphism of G -sets on each orbit. Let k_i be the restriction of k on the orbit $G * x_i$, since k is a automorphism of G -sets k_i is also a automorphism of G -sets for $G * x_i$ thus $k_i \in \text{Aut}_G(G * x_i)$. Since k doesn't permute the orbits and preserves them thus acting on each orbit separately, $k = \bar{k}_1 \circ \dots \circ \bar{k}_n = h(k_1, \dots, k_n)$. Thus h is surjective.

$\forall i, j \in \{1, \dots, n\} \ni i \neq j$. $\forall \phi \in \text{Aut}_G(G * x_i) \cap \text{Aut}_G(G * x_j)$, since $\phi \in \text{Aut}_G(G * x_i)$, $\forall x \in G * x_j$, $\bar{\phi}(x) = id_X(x)$. Similarly since $\phi \in \text{Aut}_G(G * x_j)$, $\forall x \in G * x_i$, $\bar{\phi}(x) = id_X(x)$. By definition, ϕ fixes all other orbits. The only automorphism which fixes all orbits is id_X . Thus $\phi = id_X$. $\text{Aut}_G(G * x_i) \cap \text{Aut}_G(G * x_j) = \{id_X\}$. Thus h is injective.

Thus h is an isomorphism and $\text{Aut}_G(G * x_1) \times \dots \times \text{Aut}_G(G * x_n) \cong K$

$\forall i \in 1, \dots, n$, let $H_i = G^{x_i}$. By the Orbit Stabilizer Theorem, $G * x_i \cong G/H_i$. $\text{Aut}(G/H_i) \cong N(H_i)/H_i$ by problem 3. So $\text{Aut}_G(G * x_i) \cong \text{Aut}_G(G/H_i) \cong N(H_i)/H_i$.

Thus $\ker f = K \cong N(H_1)/H_1 \times \dots \times N(H_n)/H_n$. $\square$

Problem 6a. Let X be a G -set with exactly n orbits $G * x_1, \dots, G * x_n$, and assume the stabilizers G^{x_i} are all equal. Call this subgroup of G , H . Explain why as a G -set, X is isomorphic to a disjoint union of $n = |X/G|$ copies of G/H .

Proof. By the group action homework, X is a disjoint union of orbits $G * x_1, \dots, G * x_n$ so $X = \bigsqcup_{i \in \{1, \dots, n\}} G * x_i$.

$\forall i \in \{1, \dots, n\}$, $H_i := G^{x_i}$. By the Orbit Stabilizer Theorem, $G * x_i \cong G/G^{x_i} = G/H_i$ as a G -set.

So $X \cong \bigsqcup_{i \in \{1, \dots, n\}} G/H_i$. However by assumption, $\forall i, j \in \{1, \dots, n\}$, $H_i = H_j$. So by problem 4, since the stabilizers of G/H_i which is H_i are all isomorphic, $G/H_1 \cong G/H_2 \cong \dots \cong G/H_n$. Let $H = H_1 = \dots = H_n$. Thus X is isomorphic to the disjoint union $n = |X/G|$ copies of G/H . $\square$

Problem 6b. Construct an injective homomorphism from $S_n = \text{Perm}(X/G)$ into $\text{Aut}_G(X)$.

Proof. $\forall i, j \in \{1, \dots, n\}$, since $G^{x_i} = H = G^{x_j}$, by problem 4, $G * x_i \cong G * x_j$ as G -sets. Since the orbits are isomorphic as G sets, by definition there exists an isomorphism of G sets f which maps $G * x_i$ to $G * x_j$ and an inverse isomorphism of G sets f^{-1} which maps $G * x_j$ to $G * x_i$. $a : X \rightarrow X$, where $x \in G * x_i$ $a(x) = f(x)$ and $x \in G * x_j$, $a(x) = f^{-1}(x)$, and otherwise $a(x) = id_X(x)$. a is an automorphism of G -sets because f, f^{-1} , and the identity map are automorphisms of G -sets. Thus $\exists a \in \text{Aut}_G(X)$ which swaps any two orbits.

Let $\alpha : S_n \rightarrow \text{Aut}_G(X)$, where for any two cycle $(ij), \forall i, j \in \{1, \dots, n\} \ni i \neq j$, $\alpha((ij))$ is the automorphism which swaps the two orbits. Thus $\forall p \in S_n$, $\alpha(p)$ is the automorphism which permutes the corresponding orbits. $\forall k \in \ker \alpha$, $\alpha(k) = id_X$ and no orbits are swapped, thus k is the permutation which fixes $\{1, \dots, n\}$ thus $k = e$. Thus α is injective and $S_n \subset \text{Aut}_G(X)$. $\square$

Problem 6c. Describe, with proof, $\text{Aut}_G(X)$.

Proof. By assumption, $H = G^{x_1} = \dots = G^{x_n}$. Let G/H_i be the i th copy of G/H .

By Problem 6a, X is isomorphic to the union of $|X/G|$ disjoint copies of G/H . Thus $X \cong \bigsqcup_{i \in \{1, \dots, n\}} G/H_i$.

Let N be the kernel of the homomorphism $\text{Aut}_G(X) \rightarrow S_n$. Thus $N \triangleleft \text{Aut}_G(X)$. By problem 5, $N \cong \text{Aut}(G/H_1) \times \dots \times \text{Aut}(G/H_n) \cong \text{Aut}(G/H) \times \dots \times \text{Aut}(G/H) \cong N(H)/H \times \dots \times N(H)/H = (N(H)/H)^n$.

$\forall \alpha \in \text{Aut}_G(X)$, α applies an automorphism which preserves the orbits then apply an automorphism which permutes the orbits. Thus $\alpha = \sigma \circ \phi$ where $\phi \in N$ and $\sigma \in S_n$.

$\forall \alpha \in N \cap S_n$, by definition α is a permutation of the orbits which preserves all of them thus $\forall i \in \{1, \dots, n\}$, $\alpha_*(G * x_i) = G * x_i$, the only permutation which preserves all the orbits in such a way is e . Thus $\alpha = e$. Thus $N \cap S_n = \{e\}$.

Thus there exists a bijection between $N \times S_n$ and $\text{Aut}_G(X)$ as sets.

Since $X \cong \bigsqcup_{i \in \{1, \dots, n\}} G/H_i$, $\text{Aut}_G(X) \cong \text{Aut}_G(X \cong \bigsqcup_{i \in \{1, \dots, n\}} G/H_i)$ and there exists a bijection between $N \times S_n$ and $X \cong \bigsqcup_{i \in \{1, \dots, n\}} G/H_i$.

By problem 3, $\forall \phi \in \text{Aut}(G/H)$, $\exists n \in N(H)$ s.t $\phi(xH) = n \cdot xH = xn^{-1}H$ (Note $\cdot$ denotes right action).

Since H_i is just a copy of H . $G/H = G/H_1 * = G/H_n$. Thus $\text{Aut}_G(G/H) \cong \text{Aut}_G(G/H_1) \cong * \cong \text{Aut}(G/H_n)$. Thus $\forall \phi \in \text{Aut}_G(G/H)$ let $\phi_i \in \text{Aut}_G(G/H_i)$ be the isomorphic automorphism to ϕ .

$\forall i \in \{1, \dots, n\}$, $\forall \phi_i \in \text{Aut}_G(G/H_i) = \text{Aut}_G(G/H)$. $\phi(xH) = xn^{-1}H$ for some $n \in N(H)$. Let $\sigma \in S_n$ and let $\sigma' \in \text{Aut}_G(\bigsqcup_{i \in \{1, \dots, n\}} G/H_i)$ be the corresponding automorphism to σ which permutes the copies of G/H . Let $j := \sigma(i)$. $\forall xH \in G/H_i$, $(\sigma \circ \phi_i)(xH) = \sigma(xn^{-1}H) = n\sigma(xH) = \phi_j(\sigma(xH)) = (\phi_j \circ \sigma)(xH)$ where $\phi_j \in \text{Aut}(G/H_{\sigma(i)}) = \text{Aut}_G(G/H_j)$ is the equivalent automorphism to ϕ_i for the G/H_j copy of G/H .

$\forall \sigma \in S_n$, let $\bar{\sigma} \in \text{Aut}(\bigsqcup_{i \in \{1, \dots, n\}})$ be the corresponding automorphism under the map α (Problem 6b) which permutes the copies of G/H . $\forall \phi \in N$, by problem 5, $\phi = \bar{\phi}_1 \circ \dots \circ \bar{\phi}_n$. Where $\bar{\phi}_i \in \text{Aut}_G(\bigsqcup_{i \in \{1, \dots, n\}})$ is the corresponding automorphism under the map g (Problem 5) to $\phi \in \text{Aut}_G(G/H_i)$ for all i .

$\forall i \in \{1, \dots, n\}$, let $j = \sigma(i)$. $\forall xH = xH_i \in G/H_i = G/H$, $(\bar{\sigma} \circ \phi \circ \bar{\sigma}^{-1})(\sigma(xH_i)) = \bar{\sigma}(\phi(xH_i)) = \bar{\sigma}(\bar{\phi}_i(xH_i)) = \bar{\sigma}(\phi_i(xH_i)) = \phi_j(\sigma(xH_i)) = \phi_j(xH_j)$.

Thus $C : S_n \rightarrow \text{Aut}(N)$ takes $\sigma \rightarrow [(\phi_1, \dots, \phi_n) \rightarrow (\phi_{\sigma(1)}, \dots, \phi_{\sigma(n)})]$ thus a permutation is sent to a function which takes a automorphism inside $\text{Aut}(G/H_i)$ to its corresponding automorphism inside $\text{Aut}(G/H_{\sigma(j)})$. Since $N \cong (N(H)/H)^n$, let $C' : S_n \rightarrow \text{Aut}(N)$ be the corresponding map under the isomorphism between $\text{Aut}(N) \cong \text{Aut}((N(H)/H)^n)$ thus $\sigma \rightarrow [(n_1, \dots, n_n) \rightarrow (n_{\sigma(1)}, \dots, n_{\sigma(n)})]$

Thus $\text{Aut}_G(X) \cong N \rtimes_C S_n$. Since $N \cong (N(H)/H)^n$, $\text{Aut}_G(X) \cong (N(H)/H)^n \rtimes_{C'} S_n$. □

Problem 7. Let X be a G -set with exactly n -orbits as in 6), but now assume that the stabilizers G^{x_i} and G^{x_j} ARE NOT conjugate, for any i not equal to j . Describe with proof, $\text{Aut}_G(X)$.

Proof. By problem 4, if no two stabilizers of different orbits are conjugate there exists no stabilizers of different orbits which are equal, thus no two orbits are isomorphic as G -sets. Thus $G * x_i \not\cong G * x_j$ for all $i \neq j$. $G * x_i \cong G/G^{x_i}$ and $G * x_j \cong G/G^{x_j}$ and $G/G^{x_i} \not\cong G/G^{x_j}$, as G -sets. Assume the contrary, suppose $\exists \alpha \in \text{Aut}_G(\bigcap_{i \in \{1, \dots, n\}} G/G^{x_i}) \cong \text{Aut}_G(X) \ni \alpha_*(G/G^{x_i}) = G/G^{x_j}$. Then by definition, $G/G^{x_i} \cong G/G^{x_j}$ as G -sets thus a contradiction forms and thus there exists no $\alpha \in \text{Aut}_G(\bigcap_{i \in \{1, \dots, n\}} G/G^{x_i}) \cong \text{Aut}_G(X)$ which swaps $G * x_i \cong G/G^{x_i}$ and $G * x_j \cong G/G^{x_j}$. Thus $\forall \alpha \in \text{Aut}_G(X)$, α preserves all the orbits ie. $\forall G * x \in X/G$, $\alpha_*(G * x) = G * x$. Thus $\alpha_*(G * x_i) = G * x_i$ for all i . Thus α doesn't permute any orbits and thus maps to the identity from the map $f : \text{Aut}_G(X) \rightarrow S_n$. Thus $\alpha \in \ker f = K$. Since $K \leq \text{Aut}_G(X)$ and since $K \leq \text{Aut}_G(X)$, $\text{Aut}_G(X) = K$. Let $H_i = G^{x_i}$ for all $i \in \{1, \dots, n\}$. By Problem 5, $\text{Aut}_G(X) = N(H_1)/H_1 \times \dots \times N(H_n)/H_n$. □

Problem 8a. Let X be a G -set. Let $\text{sub}(G)$ be the set of subgroups of G . This is itself a G -set, with G -acting by conjugation. Show we have a natural map of G -sets $X \rightarrow \text{sub}(G)$, sending $x \rightarrow G^x$

Proof. Let $F : X \rightarrow \text{sub}(G)$ which sends $x \rightarrow G^x$. $\forall g \in G$, $F(g * x) = G^{gx} = gG^x g^{-1} = g * F(x)$. Thus F is a homomorphism of G -sets. □

Problem 8b. Construct the map $S : X/G \rightarrow \text{sub}(G)/G$, which takes an orbit to the conjugacy class of the stabilizer of any element of the orbit.

Proof. Let $S : X/G \rightarrow \text{sub}(G)/G$ which sends $G * x \rightarrow [G^x]$ and is induced from F . $S(G * x) = S(g * x : g \in G) = \{F(g * x) : g \in G\} = \{gG^x g^{-1} : g \in G\} = [G^x]$. □

Problem 8c. Show the fibres of the composition $X \rightarrow \text{sub}(G)/G$ are unions of orbits

Proof. Let $T : X \rightarrow \text{sub}(G)/G$ where $x \rightarrow [G^x]$. $\forall x \in X$, by definition of T , $x \in T^*([G^x])$. $\forall y \in X$, if $y \in T^*([G^x])$ then $G^y \in [G^x]$. $\forall gy \in G * y$, $G^{gy} = gG^y g^{-1}$ so $G^{gy} \in [G^x]$. Thus $G * y \subseteq T^*([G^x])$. Thus since $x \in T^*([G^x])$, then $G * x \subseteq T^*([G^x])$. $\forall y \in X \ni y \notin G * x$ and $y \in T^*([G^x])$ then $G * y \subset T^*([G^x])$. Thus if a element is in $T^*([G^x])$ its entire orbit is also in $T^*([G^x])$. Thus $T^*([G^x])$ consist of a union of different orbits. $\square$

Problem 8d. Orbits are isomorphic iff they are in the same fibre.

Proof. $\implies$: Suppose $G * x \cong G * y$ as G -sets, by problem 4, G^{gx} and G^{gy} are conjugate for all $g \in G$. Thus by definition of conjugacy classes and the definition of T , $G * y \subseteq T^*([G^x])$ and $G * x \subseteq T^*([G^x])$. $\Leftarrow$: Suppose $G * x$ and $G * y$ are in the same fibre, then $G^y \in [G^x]$. By the definition of conjugacy class and problem 4, $G * x \cong G * y$ as G -sets. $\square$

Problem 8e. Describe, with proof, $\text{Aut}_G(X)$.

Proof. Let X_T be the set of all the distinct fibres of T . So $X_T = \{T^*([G^x]) : [G^x] \in \text{sub}(G)/G\} = \{T^*([G^{x_1}]), \dots, T^*([G^{x_k}])\}$ where $i \neq j$, $T^*([G^{x_i}]) \neq T^*([G^{x_j}])$.

Let $k = |X_T|$ or the number of distinct fibres of T . For $i \in \{1, \dots, k\}$, $T_i := T^*([G^{x_i}])$.

$\forall i \in \{1, \dots, k\}$, by problem 8d, T_i consists of disjoint isomorphic orbits. Let $\forall G * a, G * b \in T^*([G^x])/G$, $G * a \cong G * b$ as G -sets. By problem 4, $\exists x \in G * a$ and $\exists y \in G * y$ such that $G^x = G^y$. $G * a = G * x \cong G/G^x$ and $G * b = G * y \cong G/G^y$ and $G^x = G^y$. Thus every orbit in T_i is isomorphic to $n = |T_i/G|$ copies of G/G^{x_i} . By problem 6, $\text{Aut}_G(T_i) \cong (\text{Aut}_G(G/G^{x_i}))^n \rtimes_{C_i} S_n$ where C_i takes $\sigma \rightarrow [\phi_i \rightarrow \phi_{\sigma(i)}]$ thus a permutation is sent to a function which takes an automorphism inside $\text{Aut}(G/H_i)$ to its corresponding automorphism inside $\text{Aut}(G/H_{\sigma(j)})$. By Problem 6, $\text{Aut}_G(T_i) \cong (\text{Aut}_G(G/G^{x_i}))^n \rtimes_{C_i} S_n \cong (N(G^{x_i})/G^{x_i})^n \rtimes_{C_i} S_n$

By problem 7, $\forall \phi \in \text{Aut}_G(X)$, ϕ cannot swap two non-isomorphic orbits. Since an orbit is not isomorphic to another orbit if they are in different fibres (Problem 8d), orbits in different fibres cannot be swapped.

$\forall i \in \{1, \dots, k\}$, and $\forall \phi_i \in \text{Aut}_G(T_i)$, define $\bar{\phi}_i \in \text{Aut}_G(X)$ such that $\forall x \in T_i$, $\bar{\phi}_i(x) = \phi_i(x)$ and $\forall x \in X \setminus T_i$, $\bar{\phi}_i(x) = \text{id}_X(x) = x$

Let $P : \text{Aut}_G(T_1) \times \dots \times \text{Aut}_G(T_k) \rightarrow \text{Aut}_G(X)$ where $(\phi_1, \dots, \phi_n) \rightarrow \bar{\phi}_1 \circ \dots \circ \bar{\phi}_k$. For all $i \neq j$, $\forall \alpha \in \text{Aut}_G(T_i)$, $\beta \in \text{Aut}_G(T_j)$ and $\forall x \in X$,

1. If $x \in T_i$, then $(\bar{\alpha} \circ \bar{\beta})(x) = \alpha(x) = (\bar{\beta} \circ \bar{\alpha})(x)$
2. If $x \in T_j$, then then $(\bar{\alpha} \circ \bar{\beta})(x) = \beta(x) = (\bar{\beta} \circ \bar{\alpha})(x)$
3. If $x \in X \setminus (T_i \cup T_j)$, $(\bar{\alpha} \circ \bar{\beta})(x) = x = (\bar{\beta} \circ \bar{\alpha})(x)$

Thus $\alpha \circ \beta = \beta \circ \alpha$. Thus $\text{Aut}_G(T_i)\text{Aut}_G(T_j) = \text{Aut}_G(T_j)\text{Aut}_G(T_i)$, $\forall i, j \in \{1, \dots, k\} \ni i \neq j$. Thus P is a homomorphism because $\text{Aut}_G(T_1), \dots, \text{Aut}_G(T_k)$ all commute with each other.

$\forall \phi \in \text{Aut}_G(X)$, since ϕ cannot swap orbits in different fibres. It is only composed of automorphisms of each fibre. Thus $\phi = \bar{\phi}_1 \circ \dots \circ \bar{\phi}_k$ where $\phi_i \in \text{Aut}_G(T_i)$. Thus P is surjective.

For $\text{Aut}_G(T_i) \subseteq \text{Aut}_G(X)$ and $\text{Aut}_G(T_j) \subseteq \text{Aut}_G(X)$, $\forall \bar{\alpha} \in \text{Aut}_G(T_i) \cap \text{Aut}_G(T_j) \ni i \neq j$. Since $\bar{\alpha} \in \text{Aut}_G(T_i)$ by definition, $\forall x \in T_j$, $\bar{\alpha}(x) = \text{id}(x) = x$. The only automorphisms in $\text{Aut}_G(T_j)$ which fixes T_j is the identity. thus $\alpha = e$. Thus P is injective.

Thus P is a isomorphism and $\text{Aut}_G(T_1) \times \dots \times \text{Aut}_G(T_k) \cong \text{Aut}_G(X)$. However since $\text{Aut}_G(T_i) \cong (\text{Aut}_G(G/G^{x_i}))^{|T_i|} \rtimes_{C_i} S_n \cong (N(G^{x_i})/G^{x_i})^n \rtimes_{C_i} S_{|T_i|}$. $C_i : S_{|T_i|} \rightarrow \text{Aut}(\text{Aut}_G(G/G^{x_i})^{|T_i|})$ takes $\sigma \rightarrow [(\phi_1, \dots, \phi_{|T_i|}) \rightarrow (\phi_{\sigma(1)}, \dots, \phi_{\sigma(|T_i|)})]$ thus a permutation is sent to a function which takes a

automorphism inside $Aut(G/G^{x_i})$ to its corresponding automorphism inside $Aut(G/G^{x_{\sigma(i)}})$. Since $Aut(Aut_G(G/G^{x_i})^{|T_i|}) \cong (N(H)/H)^{|T_i|}$, let $C'_i: S_n \rightarrow (N(H)/H)^{|T_i|}$ be the corresponding map under the isomorphism between $Aut(Aut_G(G/G^{x_i})^{|T_i|}) \cong Aut((N(H)/H)^{|T_i|})$ thus $\sigma \rightarrow [(n_1, \dots, n_{|T_i|}) \rightarrow (n_{\sigma(1)}, \dots, n_{\sigma(|T_i|)})]$.

So $Aut_G(X) \cong ((N(G^{x_1})/G^{x_1})^{|T_1|} \rtimes_{C'_1} S_{|T_1|}) \times \dots \times ((N(G^{x_k})/G^{x_k})^{|T_k|} \rtimes_{C'_k} S_{|T_k|}) \quad \square$